

# STACK ATTACK

The Botball Explorer Challenge - Quick-Reference Sheet

5-MIN BRIEFING

**The mission:** The AI network running the world's smart warehouses has gone dark, and crates, pallets, and packaging have piled up in disarray. Your team's job is to design, build, and program autonomous robots that sort, stack, and deliver cargo to get the warehouse - and Botguy - back online.

## 01 THE BASICS

Match: **150 sec**

Robots: **up to 2 / team**

Field: **93" x 45"**

Start height: **15" max**

- Robots must start **fully inside** their starting box.
- Fully autonomous** once the match begins - no remote or wireless control of any kind.
- One well-placed action can trigger **several missions at once** - plan paths that stack value.

## 02 MATCH FLOW

### 150 SECONDS - FULLY AUTONOMOUS

- Teams queue, place robots, and step back - the run belongs to the code.
- Judges score missions live as crates, freight, and packaging move.
- Matches repeat across the day; best runs count toward standings.
- Discovery:** teams may touch robots while inside the start box; touching a robot once it's outside the start box is a penalty.
- Expedition:** teams have time-out cards they can use if needed during a match.

## 03 WHAT'S ON THE FIELD

### Crates & pallets

- Small crates + larger 4" crates, sorted by color/type
- Pallets sit at interior & exterior docks

### Packaging & freight

- Loose packaging material sorted into floor zones or bins
- Freight delivered and stacked to marked points

### Boundaries & Botguy

- Black tape marks zone edges - pushing pieces off it can score
- Botguy himself may need to reach a starting box

## 04 SCORING BUILDS IN STAGES

### BASE

Simple, driving-and-pushing actions - e.g. clearing a game piece off the borders, or dropping a couple of crates on a dock.

### BONUS

Sorted or multi-step moves - e.g. stacking crates, sorting packaging by color, or filling a bin correctly.

### ADV. BONUS

Sensor-driven precision - e.g. using a camera to build sorted stacks on a dock without human help.

One action can trigger more than one tier at once - the exact point values and full mission list live in the official rulebook.

## 05 DIVISIONS & ROBOTS

### DISCOVERY

#### GRADE 6 & BELOW

Teams may remove any game pieces they choose before the match. 3D-printed parts are **not permitted**.

### EXPEDITION

#### GRADES 7-12

Full field, full mission set. 3D-printed robot parts are permitted alongside kit parts.

A student of any age may always **play up** a division (e.g. a 5th grader competing in Expedition) - just not down. You don't need to buy new hardware: any robot you already own can be used, as long as it's capable of fully autonomous operation.

✓ **Spirit of Botball:** 100% student-driven - adults and mentors may not build, program, test, or document the robot. Sportsmanship, honesty, and respect toward judges, volunteers, and opponents are expected at every event.